SRIVISNU A

Machine Learning Engineer | AI Engineer | Computer Vision
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LinkedIn | GitHub | Portfolio

Professional Summary

Computer Science graduate specializing in machine learning and AI engineering, with hands-on experience building computer vision systems, deep learning pipelines, model evaluation workflows, explainable AI applications, REST APIs, and deployable ML prototypes. Skilled in Python, PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, YOLOv8, CNNs, transfer learning, Grad-CAM, Docker, GitHub Actions, Flask, SQL, and Linux.

Education

Sathyabama Institute of Science and Technology, Chennai **2022 – 2026**
B.E. Computer Science and Engineering (Data Science) *CGPA: 7.44/10.0*

Technical Skills

Programming: Python, SQL, Java **ML/AI:** PyTorch, TensorFlow/Keras, Scikit-learn, CNNs, Transfer Learning, Multi-Label Classification, Model Evaluation
Computer Vision: OpenCV, YOLOv8, FaceNet, DeepFace, OCR, Grad-CAM, DICOM, Image Preprocessing **Data:** Pandas, NumPy, Feature Engineering
MLOps/Engineering: Git, GitHub Actions, Docker, Flask, REST APIs, Linux, CI/CD, Reproducible Training

Experience

O Clock Software, Chennai **Sep 2025 – Oct 2025**
Python Developer Intern

- Developed reusable Python workflows for data processing, preprocessing, validation, and automation supporting downstream application workflows.
- Built Python scripts for recurring operations and structured data preparation, improving workflow consistency and reducing repetitive manual intervention.
- Used Git for collaborative development and worked with senior developers to troubleshoot application issues and deliver maintainable Python solutions.

Machine Learning & AI Projects

ChestXpert - Explainable Chest X-Ray AI *Python, PyTorch, ResNet-18, Grad-CAM, Flask, Streamlit, Docker, DICOM*

- Built an end-to-end multi-label chest X-ray classification pipeline using ResNet-18 transfer learning on a 2,000-image development subset with 14 abnormality labels.
- Prepared 1,582 training and 418 held-out images; implemented preprocessing, model training, per-label AUROC, Average Precision, precision, recall, and F1 evaluation.
- Implemented Grad-CAM explainability, DICOM/PNG/JPEG preprocessing, and a Flask REST API plus Streamlit review interface for model inference and visual explanations.

github.com/srivisnuA/ChestXpert-Explainable-Chest-X-Ray-AI

FaceFare - Contactless Public Transport Payment System *Python, Flask, FaceNet, DeepFace, OpenCV, SQLite, Socket.IO, Docker*

- Engineered a real-time facial-recognition fare system that reduced simulated passenger boarding time by 33% (6s to 4s per passenger) through automated identification and fare processing.
- Built a Flask backend with 12 routes, 5 Socket.IO handlers, and 8 passenger-management API endpoints; implemented SQLite wallet transactions, face enrollment, and real-time dashboard updates.
- Integrated OpenCV/FaceNet recognition with 640x480 camera input, Docker/Gunicorn deployment, and GitHub Actions CI for automated project workflows.

github.com/srivisnuA/FaceFare

SalesPilot AI - AI Sales Automation Platform *Python, FastAPI, React, Vite, Tailwind, Recharts, SQLAlchemy, OpenAI-compatible API*

- Built a full-stack AI sales automation platform connecting explainable lead scoring, personalized outreach, engagement tracking, pipeline intelligence, deal prediction, and ROI analytics.
- Implemented an explainable 0-100 lead scoring engine, engagement-driven score recomputation, pipeline forecasting, and factor-level deal risk/win explanations using a transparent heuristic approach.
- Integrated an OpenAI-compatible email generation workflow with deterministic fallback personalization, rule-based automations, execution history, and a React dashboard backed by FastAPI.

github.com/srivisnuA/salespilot-ai

Vision-Based Safety & Fraud Detection During Vehicle Refueling *Python, YOLOv8, PyTorch, CNN, OpenCV, OCR, Flask*

- Engineered a real-time vision pipeline combining 4 model inference stages for fuel-nozzle, vehicle, pose, and fire/smoke detection using YOLOv8 and PyTorch.
- Developed a Flask monitoring dashboard with a 5-state alert engine, spatial validation, fuel-flow estimation, OCR-oriented meter extraction, and real-time anomaly alerts.

github.com/srivisnuA/Vision-Based-Safety-Lapse-Detection-During-Vehicle-Refueling

Patent

FaceFare: Vision-Based Public Transport Payment System Using Deep Learning and Facial Biometrics - Indian Patent Application No. 202641031039 A, filed 15 Mar 2026, published 03 Apr 2026. Co-inventor.

Certifications

NSIC - AI Model Development using MLOps, Feb 2025 | **AWS** - Advantage Pro Certification